HOUSEHOLD FOOD WASTE PREDICTION USING MACHINE LEARNING CLASSIFICATION ALGORITHMS

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ABSTRACT

Food waste at home is affected by how people buy food, how much they serve, how they store it, and how they eat. This study uses a machine learning method to classify how much food waste households produce, based on survey data collected directly by the researcher. The survey responses were organized into records for each type of food and cleaned to create 170 unique records with 16 different pieces of information. The main thing being predicted is the *Wasted Percentage*, which was divided into four groups: less than 25%, 25–50%, 51–75%, and more than 75%. Five different machine learning methods were used—Decision Tree, Random Forest, Logistic Regression, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN)—and they were tested using a split of 80% training data and 20% testing data. The models were checked using accuracy, macro precision, macro recall, and macro F1-score. The results showed that SVM had the best accuracy at 73.53%, while Random Forest had the highest macro F1-score of 0.26. However, the dataset had a big imbalance, with most of the data falling into the lowest waste category. This made it hard for the models to accurately predict the higher waste categories. These results show that machine learning can be useful for analyzing household food waste surveys, but they also point out the need for more data, better balance between categories, and stronger testing methods to make the models more reliable.

Keywords— Food waste, primary survey, machine learning, classification, household behavior, exploratory data analysis.

I. INTRODUCTION

Food waste is something that happens at home and is connected to how people buy, prepare, measure, store, and eat food. Looking at how food waste happens can help families see what they're doing and make food use more thoughtful. Learning more about these things can also help with better food planning and reduce waste that isn't necessary. Traditional surveys give a general idea of how people behave at home. Using machine learning helps look at how different survey answers connect to types of food waste. A trained system can learn from examples and group new data into one of several groups based on what it knows. This study looks at five different machine learning methods using data collected directly from households through a survey about food waste. The main focus is on how much food is wasted, measured as a percentage, and divided into four groups: less than 25%, 25 to 50%, 51 to 75%, and more than 75%. The study covers steps like preparing the data, looking at it closely, turning data into a form the model can use, training the model, and testing how Well it works. The key points of this study are: (1) making a survey dataset collected by the researcher; (2) looking at patterns of food waste and related household habits; (3) comparing five classification methods using the same split of data for training and testing; and (4) talking about the problems of unequal data and other issues that affect how reliable the results are. The findings offer a first look at how useful machine learning can be for studying household food waste and point out where more research could be done.

II. RELATED WORK

Previous studies on household food waste have looked at factors like how people buy food, how they store it, portion sizes, and how they eat. These factors help explain why food gets wasted and show the importance of better planning and food management at home. Surveys are especially helpful for gathering information about personal habits and understanding why some people waste more food than others. Machine learning has been used in many areas to classify things by finding patterns in data and predicting which group something belongs to. Different algorithms like Decision Tree, Random Forest, Logistic Regression, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN) are used for this purpose. How well these algorithms work depends on things like how much data there is, how good the features are, how balanced the categories are, and which evaluation methods are used. Exploratory data analysis and data cleaning are important parts of building trustworthy classification models. Steps like removing duplicate records, converting categorical data into numbers, and checking how the target categories are spread out help in preparing the survey

data for analysis. In cases where some categories are underrepresented, just looking at accuracy might not tell the full story. Using metrics like macro precision, macro recall, and macro F1-score can give a better idea of how well the model handles less common categories. Taking all these into account, this study uses five different classification algorithms on primary survey data about household food waste. The performance of these algorithms is compared using a standard train-test split. The study also looks at how class imbalance affects the results and emphasizes the need for more data and better ways to test models to get more accurate predictions across different levels of food waste.

III. DATASET COLLECTION AND DESCRIPTION

A. Primary Data Collection

The dataset used in this study was gathered directly by the researcher through a survey of household food waste. It includes details about household features and the ways people buy, choose, prepare, store, eat, and waste food. The survey responses form the foundation for analyzing patterns in household food waste using machine learning techniques. The original survey data had 104 rows and covered several sections related to food. During the data cleaning process, these sections were reorganized into a uniform format that focuses on each food item. The cleaned records were then combined into one dataset. Any entries without a defined food group were removed to ensure the data remained consistent for analysis. After this step, the dataset had 178 food-level entries. Removing duplicates brought the number down to 170 unique records with 16 columns. These records represent individual food items rather than separate survey participants. The cleaned dataset was used for exploring the data, preparing features, and building machine learning models. This process helped create a consistent structure for comparing five different classification algorithms.

B. Attributes and Target Variable

The dataset includes several attributes that describe household features and how food is managed, such as age group, household size, who provided the response, the type of food, why it was bought, portion size, how many people it was meant for, how it was stored, how often it was eaten, the percentage that was consumed, the percentage that was wasted, reasons for waste, and what was done with leftovers. These features help show how food is bought, stored, eaten, and wasted in homes. The original survey also had timestamps, email addresses, and names, but these were not used in the model because they weren't needed for the task of classification. The main thing we're trying to predict is the Wasted Percentage, which shows how much food was thrown away, and it's split into four groups: less than 25%, 25% to 50%, 51% to 75%, and more than 75%. Out of 170 records, there were 113 in the lowest-waste group, 38 in the next, 12 in the third, and 7 in the highest-waste group. This shows that most of the data is from the group with the least food waste, while the higher waste groups have far fewer entries. Because the target classes are not evenly distributed, this is something to keep in mind when building and testing the model. A model might do well with the most common category but not so well with the less common ones. To get a better understanding of how well the models are performing, this study looks at accuracy, macro precision, macro recall, and macro F1-score.

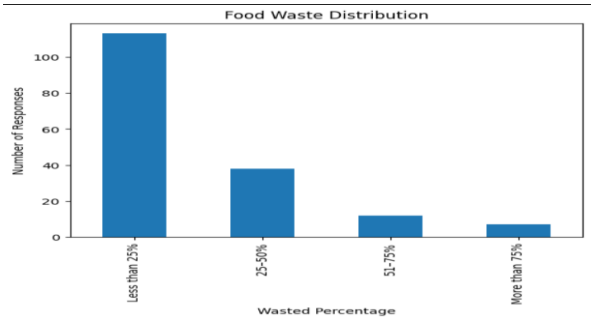


Fig. 1. Distribution of target classes in the household food waste dataset

IV. DATA PREPROCESSING AND METHODOLOGY

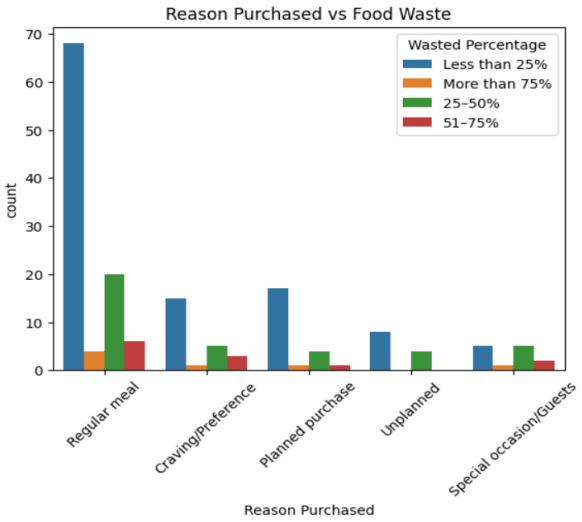
A. DATA CLEANING

The survey data was imported into a Pandas DataFrame for further processing. Several sections related to food were restructured to create a uniform food-level format and merged into one dataset. Any records that lacked valid food-group information were removed, and any duplicate rows were identified and deleted to ensure high data quality. After these steps, the final cleaned dataset had 170 unique entries and 16 columns. The notebook confirmed there were no missing values or duplicates in the cleaned data, making it ready for data exploration and machine learning model development.

B. Exploratory Data Analysis

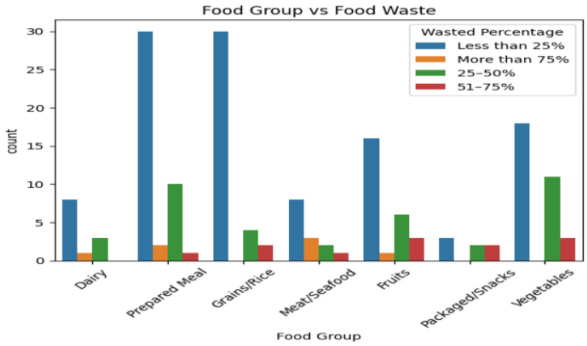
Exploratory data analysis (EDA) was conducted using Pandas, Matplotlib, and Seaborn to better understand the dataset. The notebook looked into the distribution of food waste categories and how they related to other attributes such as food group, reason for purchase, storage method, and reason for waste. These visualizations helped highlight patterns in the survey data and gave an overview before moving on to model training.

Fig. 2(a) Purchased Reason VS Food Waste



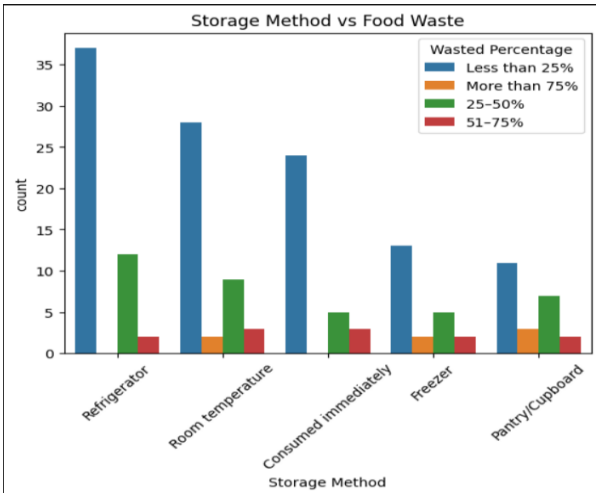
Relationship between purchase reasons and food waste levels.

Fig. 2(b) Food Group vs Food Waste



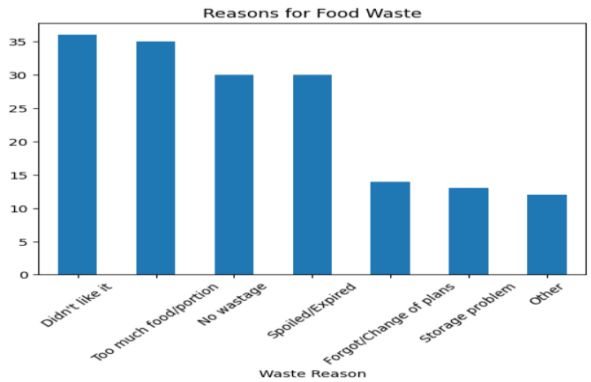
Distribution of food waste across different food groups.

Fig. 2(c) Storage Method vs Food Waste



Relationship between storage methods and food waste levels

Fig. 2(d) Reasons for Food Waste



Distribution of reported reasons for food waste

C. Feature Selection and Encoding

The target variable was set to 'Wasted Percentage'. Columns like timestamp, email address, and name were excluded since they are administrative or identifying fields. Features such as 'Consumed Percentage', 'Waste Reason', and 'Leftover Action' were also excluded to avoid using information directly linked to or potentially recorded after the waste outcome. The remaining categorical features were converted into numerical values using one-hot encoding with pandas.get_dummies(). This created a feature matrix with 170 records and 45 input features. The target vector included the corresponding 170 class labels.

D. Train-Test Split

The dataset was split into training and testing sets using the train_test_split() function with a test size of 0.2 and a random state of 42. This resulted in 136 training records and 34 testing records. The training set was used to build the classification models, while the testing set was used to assess how well they performed on new data. Since stratification was not used, the test set might not accurately reflect the original distribution of the four food-waste categories.

E. Method Overview

The experimental process included data collection from the survey, reshaping the data, cleaning it, removing duplicates, and performing exploratory data analysis. After selecting relevant features and applying one-hot encoding, the dataset was split into training and testing sets. Five classifiers were then trained and evaluated using accuracy, macro precision, macro recall, and macro F1-score to compare their performance.

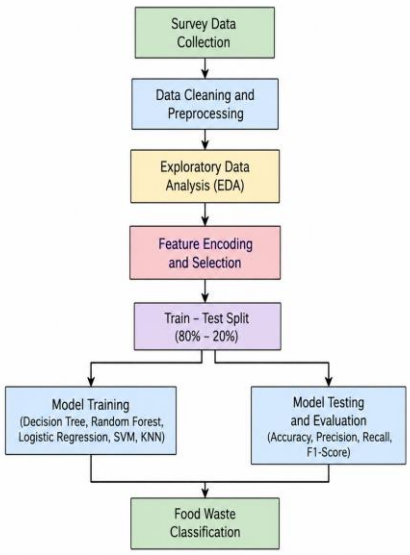


Fig. 3. Workflow of the proposed household food waste classification system.

V. CLASSIFICATION MODELS AND EVALUATION

A. Decision Tree

The Decision Tree classifier divides data into different classes based on features. It was chosen because it's easy to interpret. A fixed random state was used in the notebook to ensure the results could be reproduced.

B. Random Forest

Random Forest improves classification by combining predictions from multiple decision trees. The model used 100 trees, enabling it to consider a variety of predictions rather than relying solely on one tree.

C. Logistic Regression

Logistic Regression is a supervised classification method that estimates the relationship between input features and target classes. The model was set to run a maximum of 1,000 iterations to ensure the fitting process completed successfully.

D. Support Vector Machine (SVM)

SVM classifies data by learning decision boundaries in the feature space. It achieved the highest accuracy among the five models, but its performance was less consistent for some minority waste categories.

E. K-Nearest Neighbors (KNN)

KNN predicts the class of an observation based on the closest training examples. The notebook used five neighbors and applied Standard Scaler through a pipeline to scale features before using them for distance-based classification.

F. Evaluation Metrics

Accuracy measures the proportion of test observations that were correctly classified. Precision calculates how many predicted class labels were correct, while recall measures how many actual instances of a class were identified. The F1-score combines precision and recall into a single measure. Macro-averaged metrics calculate the score for each class separately and then take an unweighted average, giving equal importance to all four categories. Since the target classes are not evenly distributed, accuracy alone may not fully reflect a model's performance. Therefore, macro precision, macro recall, macro F1-score, and class-level reports were also considered when comparing the five classifiers.

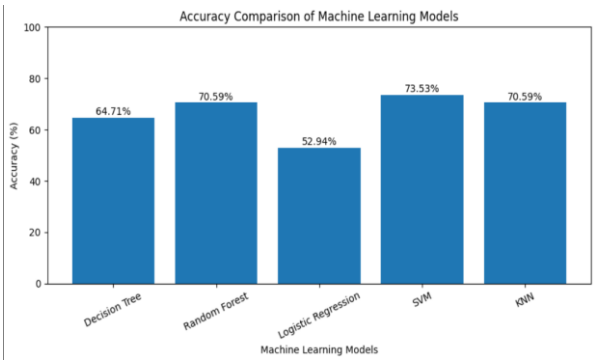


Fig. 4. Accuracy comparison of different machine learning classification models.

VI. RESULTS AND DISCUSSION

The five classifiers were tested on the same set of 34 records. SVM had the best accuracy at 73.53%, correctly identifying 25 records. Random Forest and KNN both scored 70.59% accuracy, while Logistic Regression reached 67.65% and Decision Tree scored 58.82%.

However, Random Forest had the highest macro F1-score of 0.26, followed by Logistic Regression at 0.25.

Decision Tree scored 0.23, and both SVM and KNN scored 0.21. These results show that a model with the highest accuracy doesn't always perform best when considering all four food-waste categories, especially when the dataset isn't balanced.

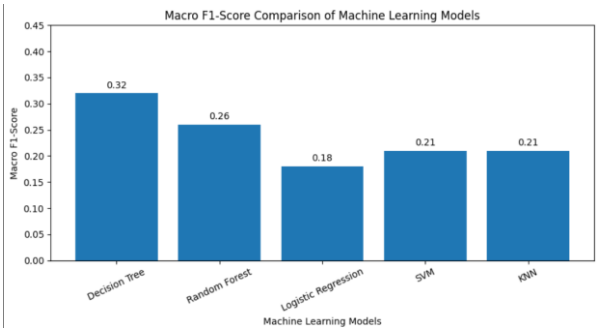


Fig. 5. Macro F1-score comparison of different machine learning models.

A. Impact of Class Imbalance

The test set had 25 records in the less-than-25% category, six in the 25–50% category, two in the 51–75% category, and one in the more-than-75% category. This uneven distribution means that correct predictions

for the most common class can greatly influence the overall accuracy, even if the model struggles with the higher-waste categories.

The class-level results show this issue. SVM had a recall of 1.00 for the less-than-25% category but didn't correctly identify any records in other categories Random Forest had a recall of 0.92 for the lowest-waste category but missed all records in the highest-waste categories. These findings show why class-level metrics are important when evaluating models on imbalanced data.

Model	Acc.	Prec.	Recall	F1
Decision Tree	64.71	0.31	0.35	0.32
Random Forest	70.59	0.25	0.27	0.26
Logistic Reg.	52.94	0.17	0.18	0.18
SVM	73.53	0.18	0.25	0.21
KNN	70.59	0.19	0.24	0.21

Table I. Performance comparison of machine learning models.

B. Interpretation

Overall, SVM had the best test accuracy, while Random Forest had the highest macro F1-score among the five classifiers.

However, the macro F1-score was still low, showing overall poor performance across all food-waste categories. These results are preliminary because the dataset is small and the categories are imbalanced. Especially, the rare categories have very few records. Collecting more survey data and using stratified validation can help check if the model rankings stay the same.

VII. LIMITATIONS

This study has several limitations. First, the dataset has only 170 food-level records, which limits the data available for training and testing Second, the target categories are highly imbalanced, with only seven records in the more-than-75% waste category. Third, the experiment used a single train-test split, so the results might vary with a different split, especially for the minority classes.

Additionally, the survey depends on self-reported practices and waste levels, which might differ between respondents and may not represent all households.

The analysis shows data patterns but doesn't show cause-and-effect relationships between household practices and food waste. Therefore, the accuracy reported should be used carefully and should not be seen as a guarantee of real-world performance.

VIII. CONCLUSION AND FUTURE WORK

This study used five supervised machine-learning algorithms on survey data collected by the researcher for household food-waste classification.

After reshaping and cleaning the data, the final dataset had 170 unique food-level records and 16 columns. The input features were converted into 45 numerical features, and the models were tested using an 80:20 train-test split.

SVM had the highest accuracy at 73.53%, while Random Forest had the best macro F1-score of 0.26.

These results highlight the importance of checking class-level performance when the target categories are imbalanced. Although this study shows the potential of machine learning for household food-waste classification, recognizing minority waste categories remains a big challenge.

In future work, the study will focus on building an AI-based food-waste prediction system by collecting a larger and more diverse dataset, balancing the waste categories, and exploring more advanced machine-learning and deep-learning techniques.

The goal is to create a reliable model that can predict household food-waste levels and help people understand waste patterns, improve food planning, and reduce unnecessary food waste. Testing and validation will be needed before applying the model in real-world household settings.

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